

Employment

Graduate Research Assistant	NC State University	April 2025 – Present
- Deployed containerized genomic platform on GCP Cloud Run (Docker, Cloud Build CI) with JBrowse 2 genome browser; enabled 100+ researchers to visualize 500GB+ datasets by eliminating on-premise infrastructure. - Improved ML model accuracy by 12% via grid-search hyperparameter sweeps and sliding-window feature engineering on multivariate biological sensor time-series.		
Product Engineer Intern	Isomer AI	May 2025 – May 2026
- Shipped 10+ production features: multi-model LLM pipelines (Claude, GPT-4o, Gemini), Temporal.io agentic workflows, and REST API integrations; 923 commits via GitHub Actions CI/CD across 7 deployments. - Owned PostgreSQL schemas, Redis caching, Auth0 RBAC, Terraform IaC, Docker, and Sentry APM; triaged and resolved production incidents by tracing failures across async microservices.		
Software Developer Intern	Mphasis (HP Company)	Jun 2024 – Aug 2024
- Improved document processing accuracy by 25% by building OCR + NLP classification pipelines that parsed unstructured health claim documents into structured fields for downstream automation. - Automated AWS and GCP releases via GitHub Actions CI/CD; configured Grafana dashboards for observability and led red-teaming to surface model reliability gaps pre-production.		
AI/ML Intern	Capgemini	Jun 2023 – Aug 2023
- Reduced ticket resolution time by 15% by training an NLP multi-class classifier that auto-routed support tickets to the correct team based on predicted category and urgency. - Cut ticket-processing latency by 20% by replacing synchronous DB polling with Kafka async ingestion feeding Spark micro-batch jobs across partitioned Hadoop HDFS.		

Education

Raleigh, NC	North Carolina State University	Aug 2024 – May 2026
- Master of Science in Computer Science - Courses: Artificial Intelligence, Operating Systems, Algorithms, Data Science, Object Oriented Development, Software Engineering.		
Mumbai, India	Mumbai University	Aug 2020 – May 2024
Bachelor of Technology in Computer Science; CGPA: 9.32/10. Courses: Data Warehousing, Big Data Infra, DS & Algorithms, Machine Learning, Blockchain.		

Projects

PulseRank	Real-Time Leaderboard & Activity Feed	2026
- Built event-driven leaderboard in Java/Spring Boot and TypeScript/NestJS: MySQL transactional writes with idempotency-key deduplication and outbox pattern for exactly-once delivery, MongoDB TTL projections, Redis sorted sets for sub-millisecond reads. - Validated multi-store consistency with Testcontainers integration tests against live MySQL, MongoDB, and Redis containers; benchmarked hot-path latency with k6 and enforced test coverage gates in GitHub Actions CI.		
VectorLift	Semantic Search & Ranking Engine	2026
- Built Kafka-driven corpus ingestion pipeline into Elasticsearch and Qdrant; served semantic search via FastAPI REST API with per-stage latency tracking across 6 configurable pipeline configurations. - Improved retrieval quality by fine-tuning bi-encoder (MS MARCO) and cross-encoder re-ranker (MiniLM); evaluated with NDCG@10, MRR@10, and paired bootstrap significance testing.		

Languages and Technologies

- Languages:** Java, Python, TypeScript, JavaScript, SQL, Bash, C
- Backend & APIs:** Spring Boot, NestJS, FastAPI, REST APIs, event-driven architecture, outbox pattern, Kafka, Temporal.io
- Data & Storage:** PostgreSQL, MySQL, Redis, MongoDB, Elasticsearch, Hadoop, Spark, Testcontainers
- Infrastructure:** Kubernetes, Docker, Helm, ArgoCD, AWS, GCP, Terraform, GitHub Actions, Prometheus, Grafana

Additional Experience and Awards

- IEEE Published:** Co-authored paper on CV-Based Cybersecurity Threat Detection (Springer icSoftComp 2023); presented to 500+ attendees.
- National Hackathon Winner:** Won or ranked Top 10 in 6+ national hackathons spanning ML, systems, and developer tools; built 3 complete MVPs end-to-end within 48-hour sprints, each with a working demo.
- Lead at Google Developers Student Club:** Led 4+ tech events (hackathons, conferences) reaching 500+ students.
- Technical Writing:** Published on LLM model collapse, AI agent security, audio AI at kulkarniaditya.com/blog